BASE CHANGE PRESERVES THE FLAT-AT-PRIMES-OVER CONDITION FOR A MODULE OVER AN ALGEBRA

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Flatness is preserved by base change and by localisation. Algebra, Lemma \ref{algebra-lemma-flat-base-change} and Algebra, Lemma \ref{algebra-lemma-flat-localization}, cited at proof lines 4100-4102. Give flatness of the localisation at the prime of the base change.

Source reference. Original lemma 052X: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6585–6585. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 052W: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 6584–6584. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HI: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 644–644. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 655–655. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

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Lemma 0.2. *Let $R \to S$ be a ring map. Let $I \subset R$ be an ideal. Let M be an S-module. Let $R \to R'$ be a ring map and $IR' \subset I' \subset R'$ an ideal. If (052W) holds for $(R \to S, I, M)$, then (052W) holds for $(R' \to S \otimes_R R', I', M \otimes_R R')$.*

Proof. Assume (052W) holds for $(R \to S, I \subset R, M)$. Let $I'(S \otimes_R R') \subset \mathfrak{q}'$ be a prime of $S \otimes_R R'$. Let $\mathfrak{q} \subset S$ be the corresponding prime of S. Then $IS \subset \mathfrak{q}$. Note that $(M \otimes_R R')_{\mathfrak{q}'}$ is a localization of the base change $M_{\mathfrak{q}} \otimes_R R'$. Hence $(M \otimes_R R')_{\mathfrak{q}'}$ is flat over R' as a localization of a flat module, see Algebra, Lemmas 00HI and 00HT. □